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ABSTRACT

Fake reviews are critical issues in the online world, as they affect the credibility of e-commerce platforms and undermine consumers' trust. Therefore, fake-review detection is of great significance. Since fake-review detection is an unsupervised problem, most existing methods and performance metrics cannot be applied. In addition, contemporary review manipulations are much more difficult to detect than before. To address these two problems, we first propose a fully unsupervised method with steps of survey research, fake-review feature analysis, fake index estimation, and fake-review selection. Fake-review features can be accurately derived from existing studies and survey research. Second, we propose a recommendation-based performance metric for evaluating fake-review detection methods. This metric differs from traditional binary classification performance metrics, as it can be used on review data with no objective review authenticity classifications. In this research, we utilize Dianping as a case study to evaluate the effectiveness of the proposed detection method and performance metric.

KEYWORDS AND PHRASES

Fake-review detection; unsupervised problem; fake reviews; online recommenders; recommender metric; recommender metric; online reviews

Online e-commerce platforms have grown in popularity over the past decade, featuring numerous consumer reviews that significantly influence consumers' purchasing decisions [9]. Reports suggest that up to 90% of consumers peruse online reviews before making a purchase, with the majority expressing confidence in the reliability of these reviews [24]. Furthermore, studies have shown that a restaurant achieving a full-star rating can increase its revenue by 5% to 9%, whereas a half-star decrease results in a 19% reduction in reservations [23]. The business value of online reviews serves as a strong motivator for some unscrupulous businesses, compelling them to manipulate these reviews, thereby undermining review credibility [13]. Such manipulations are prevalent across various online platforms, including Yelp [32], Amazon [4], Dianping, TripAdvisor [44], and others [51]. For example, Yelp discovered that approximately 25% of its reviews were manipulated, while 10.3% of Amazon's reviews were identified as fake [4, 34].

Fake reviews are inherently deceptive, as they are crafted by manipulative individuals [29]. Such reviews can potentially mislead consumers, erode consumer trust, tarnish business brands and reputations, and ultimately undermine the integrity of online purchasing [25, 46]. Therefore, it becomes critical for e-commerce platforms to develop effective detection methods for manipulated reviews. In the past, fake reviews were easily detectable because many "professional" fraudsters earned a living by generating fake reviews,

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and their activities often manifested as statistical anomalies compared with the regular reviewers' behaviors [24]. In contrast, based on our survey and existing studies, fake reviews today are much harder to spot, as they are meticulously crafted by seemingly regular reviewers and closely resemble authentic ones [4]. Furthermore, detecting fake reviews is an unsupervised problem, while most existing detection methods are designed for supervised problems [3]. Thus, fake-review detection has the following two challenges.

The first challenge is that this problem is unsupervised [24]. Fake-review detection is a problem of binary classification (i.e., "good" or "bad" identification), which can be divided into supervised and unsupervised problems [2]. Notably, most binary classification problems are supervised, while fake-review detection is unsupervised. The critical difference between these two types of problems is the presence of objective classification information. Supervised problem data have objective classification information, while unsupervised do not [15]. Loan default identification is an example of a classical supervised problem since loan default data typically include borrower classifications based on objective criteria, such as timely repayment [36]. In this dataset, borrowers who repay on time are labeled "0," while the others are marked as "1." On the contrary, data used in fake-review detection (i.e., online reviews) cannot be dualistically labeled due to the lack of a clear distinction between fake and authentic reviews, compared to on-time and overdue repayments, because it is barely possible to virtually locate the reviewers and conduct direct inquiries about the authenticity of their reviews; they will not admit the fact of giving fake reviews under manipulation or temptation even if they're interviewed. Data with objective classification information and data without objective classification information are labeled and unlabeled data, respectively [21]. In addition, unsupervised problems can only be addressed by unsupervised methods, which are few and mainly represented by clustering methods with relatively poor performance [16]. However, it is impossible to evaluate the application performance of clustering methods to unlabeled data because existing performance metrics for binary classification require labeled data [19].

The second challenge is that identifying fake reviews is more challenging today than before. On e-commerce platforms, an item (i.e., service or product) usually has numerous multimodal reviews, each of which typically consists of a rating (in the form of scoring), some text, and pictures, and the rating is essential in the post. The overall rating of an item is the average of ratings in its reviews. In our case study on Dianping, consumers tend to choose a restaurant with a high overall rating in the past. To attract consumers, unscrupulous restaurant owners hired people to publish high ratings for their restaurants and low ratings for their competitors [32]. Many "professional" fakers made a living by publishing fake reviews [51]. They were generally required to give lots of reviews only with ratings in concentrated periods, resulting in these fake reviews being easily identified compared with genuine ones [24]. Such review manipulations lead to similarly high ratings for more and more online items. According to our Dianping data in December 2022, 83.24 percent of restaurants in Beijing have the highest 5-star overall rating, making it difficult for consumers to choose from based on these same high ratings. The findings from our surveyed research (i.e., Section 1.3 of Supplement) reveal that more than 50 percent of surveyed consumers pay more attention to the content of reviews of alternative restaurants, such as text and pictures, than to their overall ratings. Accordingly, unscrupulous restaurant owners occasionally hire and manipulate ordinary reviewers to post their "carefully edited" reviews with rich and

attractive content, and these reviews are harder to identify compared to previous fake reviews generated by professional fake reviewers since they look very realistic and do not seem out of place next to authentic ones [4, 24].

To address the first challenge, we develop a novel unsupervised fake-review detection method and propose a new performance metric for this unsupervised problem. The proposed method has four key steps: survey research, fake-review feature analysis, fake index estimation, and fake-review selection. Survey research is mainly used to address the second challenge, as the main participants related to reviews have the best understanding of the fake-review features. Therefore, we first conduct survey research among them to obtain fake-review features prevalent today. Then we develop a detection process to estimate the fake index of each review by integrating fake-review features identified in existing studies with those analyzed through our survey results. A higher fake index of a review implies a higher probability of being fake. We finally rank all reviews based on their fake indexes; the top ones are labeled as fake. In this research, we use Dianping in the case study to describe the application and evaluate the effectiveness of this detection method. For the proposed performance metric, the idea of personalized recommendations is adopted in evaluating detection performance. Based on reviews, consumer preference analysis is the key to personalized recommendations [12]. However, as fake reviews are manipulated and therefore cannot reflect consumers' true preferences, they badly affect recommendation performance [10, 17]. However, such performance can be improved by screening and removing detected fake reviews following the application of an effective detection method [14]. Based on this idea, we propose to use the improvement in recommendation performance as the performance metric for evaluating the effectiveness of fake-review detection methods. This performance metric can be used on review data since it does not require data with objective classifications.

In conclusion, the major contributions of this work are summarized in the following:

- (1) We propose to derive fake-review features through existing studies and surveys conducted among the prominent participants in the review process, including platform staff, reviewers, and consumers.
- (2) We explore innovative methods of quantifying and combining the derived fake-review features into a detection process to estimate review fake index and label potential fake reviews based on their fake indexes.
- (3) We propose using the improvement in recommendation performance as a novel performance metric to evaluate the effectiveness of fake-review detection methods, which can be used on online reviews for which the authenticities cannot be objectively confirmed.
- (4) We collect public reviews of restaurants in Beijing from Dianping's official website in December 2022 to construct a Dianping dataset. This dataset includes 2,065,276,933 reviews of 31,182 restaurants provided by 1,831,582 reviewers.

The remainder of the article is organized as follows: The second section reviews the relevant literature. The third section presents the proposed fake-review detection method. The fourth section develops a novel recommendation-based performance metric. The fifth section furnishes the experimental results. The sixth section concludes the study and its potential limitations.

Literature Review

Online reviews are essential for consumers' purchasing decision making and are crucial in determining business success [32]. However, some unscrupulous merchants manipulate reviews to promote themselves or denounce their competitors' products and services for more benefits [26]. Detecting such manipulated fake reviews is a serious and significant problem for both online platforms and consumers [4]. Machine learning has become the dominant approach for detecting fake reviews [22], and such practices include applying statistical learning techniques to automatically extract hidden knowledge or patterns from training data or previous experience [18, 39]. Machine learning methods are generally classified into supervised and unsupervised learning methods based on the existence of objective classification information [20].

Most existing fake-review detection methods are extensions of supervised learning methods, which require data with objective classifications. In comparison, only a few studies on fake-review detection have used unsupervised learning methods. Related literature is summarized in Table 1. Performance metrics used to evaluate these detection methods primarily include precision, recall, *F*-score, accuracy, and AUC (area under the curve) [2], given that fake-review detection is a binary classification problem. Precision is the proportion of confirmed fake reviews in the detected fake reviews, while recall is the proportion of correctly identified fake reviews among all fake reviews. A trade-off exists between precision and recall, leading to the development of an aggregate performance measure, the *F*-score, defined as the harmonic mean of precision and recall [42]. Accuracy is the percentage of correctly detected fake and authentic reviews among the total number of reviews. Notably, existing metrics for evaluating the performance of fake-review detection, whether for supervised or unsupervised learning methods, require data with classifications.

However, the problem of fake-review detection is unsupervised, as the truthfulness of a review cannot be objectively confirmed. Existing fake-review detection methods based on

Table 1. Summary of Literature on Fake-Review Detection Based on Supervised and Unsupervised Learning Methods

Method classification	Authors	Methodology
Supervised learning methods	Kumar et al. (2018) [23]	A hierarchical supervised-learning method is proposed to detect fake reviews by modeling reviewers' features in distributions and stacking these distributions using supervised-learning techniques.
	Zhang et al. (2022) [52]	Manipulated reviewers are identified from handling data imbalance based on weighted latent Dirichlet allocation and Kullback–Leibler (KL) divergence.
	Xu et al. (2019) [50]	The hierarchical dual-attention recurrent neural network is used for sentiment analysis to calculate fake levels of reviews.
	Satia et al. (2021) [43]	A set of 133 distinctive features is derived from combining content and behavior-based attributes to identify fake reviews through deep machine learning classifiers.
	Lau et al. (2012) [27]	A novel text-mining model based on support vector machines is developed and integrated into a semantic language model to detect deceptive reviews.
Unsupervised learning methods	Kuma et al. (2019) [24]	A fully unsupervised approach using mixture methods is presented to detect manipulation behaviors in online reviewers, combining univariate and multivariate distributions into a unified anomaly detection method.
	Mukherjee et al. (2013) [33]	Clustering methods are utilized to identify fake reviewers, creating a distributional divergence between the latent population distributions of fake and authentic reviewers.
	Liang et al. (2016) [30]	Unsupervised text-mining techniques are used to identify common features of many malicious insiders.

supervised learning require data with objective classifications, and are thus unsuitable for this problem. Even unsupervised learning methods require classified data to test their performance. Therefore, existing binary classification performance metrics cannot be used on fake-review detection data. Furthermore, the current fake reviews are hard to identify since they seem genuine. The detection method and performance metric proposed in this research aim to overcome these challenges.

Fake-Review Detection Method

Analysis of Fake-Review Features

Since a rating is usually included along with accompanying text or pictures in a typical review, we analyze the features of ratings, text, and pictures in fake reviews based on existing studies and our survey research.

Analysis of Fake-Review Features Based on Existing Studies

Based on existing studies, manipulated reviewers are often instructed to post extreme ratings in a five-point rating system, that is, “5” or “1,” for their own business or competitors, respectively [31, 32, 49]. Content duplication is a prominent feature of fake reviews for the text, and some fake reviews show similar writing styles [23, 24, 27]. For pictures, the likelihood of review manipulation is significantly increased when multiple pictures are identical in different reviews [45], as it is implausible for different reviewers to capture identical images.

Analysis of Fake-Review Features Based on Survey Research

In our case study on Dianping, we conducted interviews with Dianping staff (Interview) and questionnaire surveys with reviewers and consumers. Questionnaire 1, consisting of six questions (QR1, QR2, QR3, QR4, QR5, and QR6), is presented in Section 1.2 of the Supplement and is intended for reviewers. Questionnaire 2, containing eight questions (QC1, QC2, QC3, QC4, QC5, QC6, QC7, and QC8), is provided in Section 1.3 of the Supplement and is designed for consumers. Due to space limitations, the specific details of the survey research can be found in Supplement 1, while the research conclusions and results are presented in the following content.

The rapid growth of Dianping has fostered consumer trust and reliance on its reviews, particularly in restaurant choices (QC1, QC2, and QC7). As most restaurant ratings on Dianping are now high (Interview), consumers have shifted their attention to the review content, including text and images, rather than focusing solely on the ratings (QC4). Consumers generally read only a few top reviews in the default sorting for each option (QC5 and QC6); these reviews largely influence their decision making (QC3). Considering the importance of default review sorting, the default review sorting on Dianping is based on their quality rather than their timestamp (Interview).

With these changes in mind, it is evident that the landscape of review manipulation has also evolved. On Dianping, the ranking of a review is determined through a comprehensive assessment of its reviewer level, ranging from Lv. 1 to Lv. 8 (from the lowest to the highest

level), as well as the richness of the content (Interview). Unscrupulous restaurant owners now employ and manipulate high-level reviewers, typically Lv. 5 to Lv. 8, to generate reviews with lengthy text and appealing pictures (QR5). This strategy allows these manipulated fake reviews to easily secure top positions and capture consumers' attention. The reviewer's level is closely related to the likelihood of their participation in fake reviews, where nearly 90 percent of Lv. 8 reviewers have posted at least one fake review on Dianping (QR1 and QR2). However, it is essential to note that most of their reviews are authentic (QR3), which poses a significant challenge in identifying fake reviews. Therefore, current fake reviewers are not as "professional" as their predecessors [24], which means prior fake-review detection methods are ineffective now. Furthermore, although there are fewer manipulated reviews on Dianping nowadays, the issue of fake reviews has become more severe. The reason is that consumers tend to read a limited number of top reviews (QC5), a considerable proportion of which are fake.

Nevertheless, new review manipulations also bring in new fake-review features. We have identified these novel fake-review features in terms of ratings, text, images, and reviewer level, which are outlined in the following (Questionnaire 1):

- (1) Fake reviews generally include the highest rating (e.g., five stars).
- (2) Some assessed dishes are identical in different fake reviews of the same restaurant.
- (3) Some images are identical in different fake reviews of the same restaurant.
- (4) Review manipulation is commonly observed among higher level reviewers, particularly Lv. 5 to 8.

It should be noted that reviews with these features are not necessarily fake; however, fake reviews share such features. Besides, recognizing all the features of fake reviews can be challenging. Therefore, we must acknowledge that our analysis, drawn from existing studies and survey research, only encompasses certain features of fake reviews, presenting a limitation in this research.

Fake-Index Estimation

These fake-review features can be utilized to develop a feature-based fake-review detection process. Denote by $\mathbf{Rr} = \{Rr_i \mid i \in \{1, \dots, m\}\}$ the set of reviewers and $\mathbf{Rt} = \{Rt_j \mid j \in \{1, \dots, n\}\}$ as the set of restaurants. Then, let $\mathbf{Rw} \subset \mathbf{Rr} \times \mathbf{Rt}$ be the review set, where $Rw_{i,j}$ represents the review of Rt_j published by Rr_i . Each review $Rw_{i,j} = \{r_{i,j}, t_{i,j}, p_{i,j}\}$ includes a rating ($r_{i,j}$), text ($t_{i,j}$), and pictures ($p_{i,j}$). Denote by $FI_{i,j}$ the fake index of the review $Rw_{i,j}$. We estimate a fake index for each review element (i.e., $r_{i,j}, t_{i,j}, p_{i,j}$) by quantifying its features and combining the three indexes into a comprehensive fake index (FI).

Fake-Index Estimation Based on Picture Features

According to previous research, having identical pictures in different reviews is a significant fake-review feature since it is naturally impossible for different people to take identical pictures of the same target [45]. Figure 1 shows an actual example excerpted from Dianping, where three pictures are the same in these two reviews; hence, they are highly likely to be manipulated. We estimate the fake index from the



Figure 1. An Example of Different Reviews With the Same Pictures

picture feature based on the number of identical pictures in different reviews of a restaurant. Structural similarity ($SSim$) is introduced to measure the picture similarity and falls into the interval of $[0, 1]$ [48]. The higher the $SSim$ of the two pictures, the higher is their similarity. Two pictures are identical when $SSim$ equals 1. Denote by $Rw_j = \{Rw_{i,j} \mid i \in \{1, \dots, m\} \cap p_{i,j} \neq \emptyset\}$ the set of reviews to Rt_j containing pictures. Let $p_{i,j} = \{p_{i,j}^k \mid k \in \{1, \dots, |p_{i,j}|\}\}$ be the picture subset of $Rw_{i,j}$. For two reviews, $Rw_{p,j}$ and $Rw_{q,j}$, the number of the same pictures in them is

$$T(SSim) = \begin{cases} 1 & SSim = 1 \\ 0 & SSim \neq 1 \end{cases} \quad (1)$$

$$N_{p,q} = \sum_{a=1}^{|p_{p,j}|} \sum_{b=1}^{|p_{q,j}|} T(SSim_{p_{p,j}^a, p_{q,j}^b}) \quad (2)$$

where $T(x)$ is the sign function, with a value of 1 only when x equals 1.

For two reviews, $Rw_{p,j}$ and $Rw_{q,j}$, the number of the same pictures in them is calculated by comparing the pictures in $Rw_{p,j}$ (i.e., $p_{p,j}$) with those in $Rw_{q,j}$ (i.e., $p_{q,j}$) one by one. For m reviews of restaurant Rt_j , we can obtain an $m \times m$ symmetrical matrix, as shown in Table 2. In this matrix, diagonal elements are set to 0. In addition, if either Rr_p or Rr_q does not publish reviews to Rt_j or either of these two has no pictures, $N_{p,q}$ is set to 0. For all restaurants on the platform (i.e., $Rt = \{Rt_j \mid j \in \{1, \dots, n\}\}$), we can obtain n such matrices. Finally, the fake index of $Rw_{i,j}$ from picture features can be estimated as

Table 2. The Matrix of the Same Pictures Between Reviews of Rt_j

	$Rw_{1,j}$	...	$Rw_{p,j}$	$Rw_{q,j}$	...	$Rw_{m,j}$
$Rw_{1,j}$	0	...	$N_{1,p}$	$N_{1,q}$	...	$N_{1,m}$
...	...	0	...	...	...	...
$Rw_{p,j}$	$N_{p,1}$	...	0	$N_{p,q}$	...	$N_{p,m}$
$Rw_{q,j}$	$N_{q,1}$	...	$N_{q,p}$	0	...	$N_{q,m}$
...	...	...	...	...	0	...
$Rw_{m,j}$	$N_{m,1}$	...	$N_{m,p}$	$N_{m,q}$	...	0

$$FI_{i,j}^{\text{pic}} = \sum_m^{k=1} N_{i,k}. \quad (3)$$

Fake-Index Estimation Based on Text Features

According to survey results, manipulated reviews tend to focus on specific dishes at a restaurant, and the comments used to compliment these dishes are often similar across different reviews. To highlight the dishes in the text, reviewers prefer to use particular writing styles, such as specific punctuation, as shown in Figure 2. Based on the special writing styles in reviews of Rt_j , first, k dishes evaluated in these reviews are identified by text segmentation techniques [41], and an Rt_j 's dish set, $\mathbf{D}_j = \{d_k \mid k \in \{1, \dots, u\}\}$, is formed. Second, a matrix is formed based on $\mathbf{D}_j$ and evaluated dishes in different Rt_j 's reviews, as shown in Table 3.

In this matrix, “1” indicates that the corresponding dish (e.g., d_k) is evaluated in the corresponding review (e.g., $Rw_{1,j}$), while “0” indicates that it is not evaluated. Accordingly, the number of elements in the intersection of two sets is used to estimate the dish similarity of $Rw_{p,j}$ and $Rw_{q,j}$ as follows:

$$Jc_{p,q}^D = |D_{p,j} \cap D_{q,j}| \quad (4)$$

**Figure 2.** Example of Specific Writing Styles in a Review

Table 3. Matrix of Dishes in Reviews of Rt_j

	d_1	...	d_k	...	d_u
$Rw_{1,j}$	0	...	<u>1</u>	...	1
...	...	...	...	...	...
$Rw_{p,j}$	0	...	0	...	0
$Rw_{q,j}$	1	...	1	...	1
...	...	...	...	...	...
$Rw_{m,j}$	1	...	0	...	0

where $|A|$ is the number of elements in set A . $D_{p,j}$ and $D_{q,j}$ are two evaluated dish sets of $Rw_{p,j}$ and $Rw_{q,j}$, respectively. $Jc_{p,q}^D$ represents the number of identical evaluated dishes in these two reviews. In addition, considering that the content similarity is a classical fake-review feature [27, 51], Word2vec is adopted to estimate the evaluation content similarity between $t_{p,j}$ and $t_{q,j}$ [8], denoted as $Cs_{p,q}^T$.

In the context of fake-review detection, text similarity should be assessed based on both evaluated dish similarity (i.e., Jc) and evaluation content similarity (i.e., Cs). Our survey research has shown that unscrupulous restaurant owners may invite reviewers to enjoy complimentary meals and, in return, require them to post positive reviews, especially for the dishes offered. As these invited reviewers experienced the same dishes, the evaluated dishes in their manipulated reviews tend to be similar. Additionally, to avoid antifraud detection by the platform, the owner refrains from providing specific praises for the dishes, leaving it to the reviewers to craft their specific comments. Nevertheless, as previous studies have indicated that similarity in evaluation content is a feature of fake reviews [51], we assert that evaluation content similarity should not be neglected but should, instead, assume a secondary role compared to the similarity in evaluated dishes. Therefore, we adopt evaluated dish similarity (i.e., Jc) as the “main factor” and evaluation content similarity (i.e., Cs) as the “enhancement factor,” combining them in an “enhanced” manner as follows [37]:

$$Ts_{p,q}^t = Jc_{p,q}^D * \left(1 + Cs_{p,q}^T\right) \quad (5)$$

where $Ts_{p,q}^t$ is the text similarity of review $Rw_{p,j}$ and $Rw_{q,j}$. Jc is the number of identical dishes of these two reviews, a positive integer greater than 0, while the interval of Cs is $[0, 1]$. Equation 5, designed considering their value ranges, ensures that Ts is primarily influenced by Jc , with Cs as a regulating and augmenting factor. To further discuss the use of this equation in the proposed fake-review detection method, a comparative experiment is conducted in the fifth section of this article.

Based on the preceding calculations, we can obtain a similarity matrix of Rt_j 's reviews from a textual perspective, similar to Table 2. Then the fake index of a review (e.g., $Rw_{i,j}$) from its text features is

$$FI_{i,j}^{text} = \sum_m^{p=1} Ts_{p,q}^t \quad (6)$$

For convenience in the following calculations, $FI_{i,j}^{text}$ is normalized into an interval of $[0, 1]$.

Fake-Index Estimation Based on Reviewer Features

E-commerce platforms usually incentivize consumers to post reviews, hoping that such reviews will help consumers make better-informed purchasing decisions, thus increasing their reliance on the platform. For example, Dianping rewards reviewers “contribution points” for their reviews depending on content richness. In detail, a reviewer will receive only 5 points after providing a rating-only review. However, a maximum of 20 points will be granted after providing a review with rich pictures and text. Reviewers’ contribution points determine their levels from Lv. 1 to Lv. 8 in ascending order. Higher level reviewers will be offered more benefits by Dianping, such as free meals and higher discount coupons, which serve as an incentive mechanism of Dianping, motivating reviewers to post reviews in return for these benefits.

Survey results suggest that restaurant owners will likely hire high-level reviewers to post fake reviews (QR1 and QR2 of Supplement 1.2), as their reviews are usually displayed in a prominent position that is more visible to consumers. L_i is used to denote the reviewer level of Rr_i , a positive integer within the range of 1 to 8.

Fake-Index Estimation Based on the Extracted Fake-Review Features

As illustrated in Figure 3, a typical review includes a rating, text, images, and relevant information about the reviewer, primarily their levels. In the preceding sections, we have determined fake indices of reviews based on the features of pictures, text, and reviewer levels in fake reviews, denoted as $FI_{i,j}^{pic}$, $FI_{i,j}^{text}$ and L_i , respectively. Regarding the rating features of fake reviews, our survey research findings (Supplement 1.1 and QR6 of Supplement 1.2) indicate that contemporary restaurant owners tend to encourage manipulated reviewers to give their restaurants the highest ratings but do not necessarily require them to give low ratings to their competitors as in the past [26]. Consequently, a distinguishing feature of a fake review is the presence of consistently high ratings, but not all reviews with high ratings are necessarily fake. The fake index of the review $Rw_{i,j}$, based on rating features, is represented as $FI_{i,j}^{rating}$, which is normalized within the interval $[0, 1]$ by normalizing $r_{i,j}$. With these fake-review features extracted from pictures, text, ratings, and reviewer levels, the fake index of a review $Rw_{i,j}$ is estimated as follows:



Figure 3. A Review Sample

$$FI_{i,j} = L_i * \left(1 + FI_{i,j}^{pic}\right) * \left(1 + FI_{i,j}^{text}\right) * \left(1 + FI_{i,j}^{rating}\right) . \quad (7)$$

Notably, we must acknowledge that this equation cannot encompass all the features of fake reviews, even though we have meticulously analyzed the characteristics of fake reviews by considering most elements within. Identifying all the features of fake reviews is challenging, which is also a limitation of this study.

In addition, we add 1 to each term, except for L_i , on the right-hand side of this equation. This addition on $FI_{i,j}^{pic}$ plays a different role than those on $FI_{i,j}^{text}$ and $FI_{i,j}^{rating}$. The operations on $FI_{i,j}^{text}$ and $FI_{i,j}^{rating}$ aim to realize their roles as “enhancement” factors for which the values range from 0 to 1, since similar text and identical high scores in different reviews do not necessarily mean that they are fake [26]. On the contrary, since different people cannot take identical pictures of the same target, “identical pictures in different reviews” can be used as the golden standard to detect fake reviews. Therefore, $FI_{i,j}^{pic}$ deserves a more considerable weight than $FI_{i,j}^{text}$ and $FI_{i,j}^{rating}$ in the fake index estimation without normalization. The operation of adding 1 to $FI_{i,j}^{pic}$ is to ensure that FI of the review including no pictures (i.e., $FI_{i,j}^{pic}$ equals to 0) is not equal to 0. Similarly, L_i represents reviewer Rr_i ’s level, which is a positive integer greater than 1, and it is not normalized due to its significant role in review manipulation according to survey results.

Fake-Review Detection

The comprehensive fake index (i.e., FI) measures the extent of fake reviews. A higher value of FI indicates that the corresponding review is more likely to be fake. Therefore, we rank reviews in descending order based on their FI and identify Top- $N\%$, ranging from 0 percent to 100 percent, of them as fake reviews. We propose three strategies for ranking and selecting fake reviews and test their effectiveness in the fifth section.

- (1) Reviewer-oriented selection ($FRD-Rr$): For all reviewers, we select the Top- $N\%$ of their reviews as fake reviews, respectively.
- (2) Restaurant-oriented selection ($FRD-Rt$): For all restaurants, we select Top- $N\%$ of their reviews as fake reviews, respectively.
- (3) Normal selection ($FRD-R$): We select Top- $N\%$ of all reviews as fake reviews.

Figure 4 summarizes our proposed fake-review detection method, which can be used on any platform with reviews. Based on the proposed method, a detection process can be developed only based on the fake-review features identified in the existing studies without conducting surveys. Although this detection process is more generalized, its effectiveness may not be as good as the detection process based on both existing studies and survey results. With the survey research in the proposed method, the main ideas and fundamental techniques of this method are still highly generalized, which include questionnaire design in survey research, fake-review feature extraction and quantification with data mining techniques, fake index estimation considering multiple review features, and so on. In this research, Dianping is used as a case study to evaluate the effectiveness of our proposed method. Therefore, specific fake index calculations used in this case may not be applicable or duplicated to

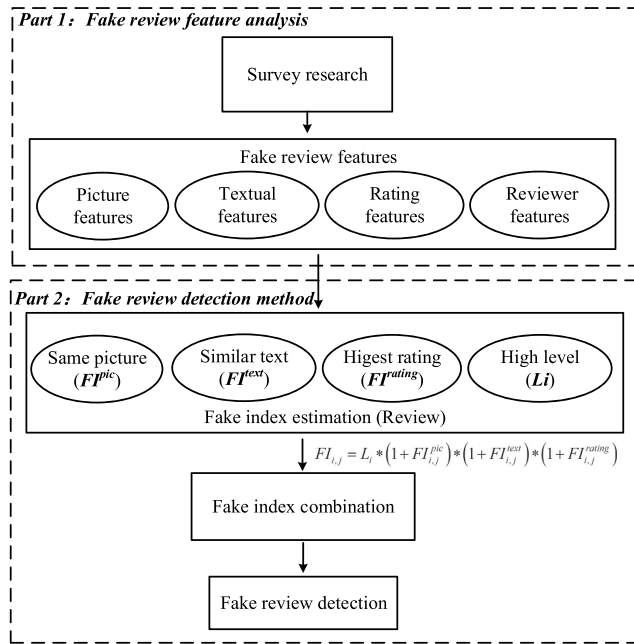


Figure 4. The Framework of the Proposed Fake-Review Detection Method

other platforms, as they are based on a combination of existing studies and Dianping survey results. When promoting the methodology to other platforms, the characteristics of the platforms should be considered in survey designs, and some specific questions may also need to be adjusted.

Recommendation-Based Performance Metric

The first challenge of fake-review detection is that this problem is unsupervised in binary classifications, as fake reviews have no objective classification information on their authenticities. Therefore, most existing classical binary classification performance metrics, such as AUC, F -measure, and accuracy, designed for supervised problems and requiring data with objective classifications [47], cannot be used for fake-review detections. To address this problem, we introduce personalized recommendations into the performance evaluation of fake-review detection and propose a novel performance metric.

Recommendation algorithms aim to suggest products or services to a target consumer based on their preferences, which are analyzed from the previous purchase and evaluation behaviors and those of other consumers with high preference similarities [35]. The process involves estimating the preference similarities between the target consumer and others, generally based on the reviews of co-used services, and falls within the interval of $[0, 1]$ [6]. A higher similarity value indicates that the preferences between two consumers are more similar [1]. The other consumers are then ranked in descending order based on their preference similarities, and the top ones are selected as the target consumer's neighbors. The services used by the neighbors but not the target consumer are regarded as the alternative services or the latter. The recommendation indexes of

these alternatives for the target consumer are calculated based on reviews and preference similarities of their neighbors. Finally, the alternative services are ranked in descending order based on their recommendation indexes, and the top 3–5 are recommended to the target consumer.

The personalized recommendation relies heavily on the accuracy of preference similarity estimation based on the reviews of co-used items. This estimation plays a significant role in the entire recommendation process, from neighbor selection to fake index calculation [28]. Consequently, the quality of preference similarity estimation greatly affects the recommendation performance [37]. However, manipulated reviews, which do not reflect true consumer preference, lead to inaccurate preference similarity results and negative impacts on the final recommendation performance [10]. The fake-review problem is a well-known challenge in recommendations [26, 50], where existing studies have shown that removing fake reviews improves recommendation performance [14, 17]. Therefore, we propose to use the improvement in recommendation performance as the performance metric for evaluating the effectiveness of fake-review detection methods. The proposed metric can be applied to review data without objective classifications.

The effectiveness of the fake-review detection method is evaluated by comparing the recommendation performance on the original dataset with that on a “clean” dataset where detected fake reviews are removed. Specifically, the effectiveness is evaluated by measuring the improvement of the recommendation performance, which is positively correlated. Rating-based recommendation methods are used to evaluate the detection performance, as reviews always contain ratings. To ensure the reliability of the performance metric, we combine the results of three classical and widely used collaborative filtering (CF) recommendation methods, including user-based CF (U-CF) [53], item-based CF (I-CF) [38], and hybrid-based CF (H-CF). The training set comprises 80% of the reviews, and the remaining data are used to evaluate the recommendation performance through mean absolute error (MAE) [1]. To reduce uncertainty, 10-fold cross-validation (10-CV) is used to obtain the MAE results. The detection performance evaluated by U-CF is

$$P_{U-CF} = \frac{MAE_{OR}^{U-CF} - MAE_{FD}^{U-CF}}{MAE_{OR}^{U-CF}} . \quad (8)$$

The recommendation performance of U-CF on the original and filtered datasets is represented by MAE_{OR}^{U-CF} and MAE_{FD}^{U-CF} , respectively. MAE falls within the $[0, 1]$ range, where a smaller value indicates better recommendation performance. Equation 8 measures the recommendation performance improvement. Similarly, we can obtain P_{I-CF} and P_{H-CF} . Finally, we calculate the average of P_{U-CF} , P_{I-CF} , and P_{H-CF} as the detection performance metric, which is defined as follows:

$$P = \frac{P_{U-CF} + P_{I-CF} + P_{H-CF}}{3} . \quad (9)$$

As MAE_{FD} falls into the interval of $[0, 1]$, the P value ranges from $\frac{MAE_{OR}-1}{MAE_{OR}}$ to 100 percent. It should be noted that $\frac{MAE_{OR}-1}{MAE_{OR}}$ can be less than 0, indicating that the recommendation performance is worse than when the review data is not filtered. In this case, many of the detection results are incorrect. Therefore, a method can only

be considered effective for fake-review detection if its P value is greater than 0. The proposed performance metric can be applied to evaluate and compare the performance of any detection method on any review data.

Experiment Studies

This research makes two main contributions in the fields of performance metrics and detection methods. To evaluate these contributions, the experiments are divided into three parts.

In the first part of these experiments, the feasibility of the proposed performance metric is assessed with Yelp data. The performance metric is considered effective if removing manipulated reviews from a dataset enhances recommendation performance on this data set and this enhancement monotonically increases with the increase of removed fake reviews. Since only Yelp labels fake reviews on its online platform, we use the Yelp dataset for this experiment.

In the second and third parts of the experiments, the proposed unsupervised fake-review detection method (*FRD*) is tested on the Dianping dataset. Specifically, the performance of *FRD* with existing well-performing detection methods is tested first. Then the proposed detection method is further analyzed by exploring the optimal setting of the Top- $N\%$ in *FRD*, comparing the *FRD* performance based on different rankings for selecting fake reviews, profoundly studying the effects of different fake-review features on fake-review detection performance of *FRD*, and discussing text similarity estimations in proposed detection methods. Due to space limitations, details of Dianping and Yelp datasets are described in Supplement 2.

Performance Metric Experiment

We conduct experiments using the Yelp dataset to evaluate the validity of the proposed performance metric. Specifically, we investigate whether removing fake reviews improves recommendation performance and whether the improvement monotonically increases with removing fake reviews.

First, we use U-CF, I-CF, and H-CF to perform personalized recommendations on both the original data (i.e., 58,517 reviews) and the filtered data (i.e., 58,517 – 8,368 ratings) and calculate MAE and P as shown in Table 4. The performance of these three recommendation methods significantly improved by approximately 30 percent (i.e., P) when all fake ratings were removed, as fake reviews adversely affect

Table 4. Performance of Recommendation Methods on the Yelp Dataset

Methods Metric	U-CF	I-CF	H-CF
MAE_{OR}	0.672	0.601	0.545
MAE_{FD}	0.845	0.836	0.782
P	31.24%	28.11%	30.31%
$P_{average}$	29.83%		

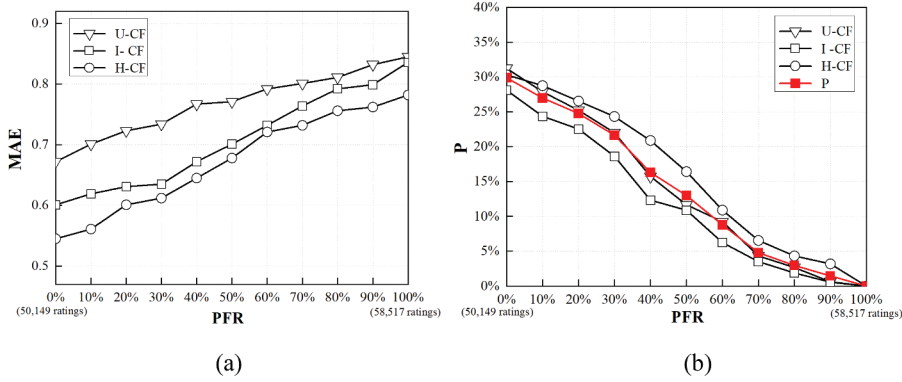


Figure 5. Performance of Recommendation Methods on Data With Different Numbers of Fake Reviews

consumer preference estimation and lead to unsatisfactory recommendations [10]. Thus, if a detection method can accurately filter fake reviews, the performance of the recommendation algorithms on the filtered data is better than that on the original data.

Second, we conduct further experiments to test the reliability of the recommendation performance. In Figure 5, the horizontal axis represents the percentage of fake reviews (PFR). For example, $PFR = 10\%$ means that the dataset contains 50,149 authentic reviews and 836 (i.e., $8,368 \times 10\%$) fake reviews randomly abstracted from 8,368 fake reviews. As shown in Figure 5(a), MAEs of U-CF, I-CF, and H-CF monotonically increase with the increase of PFR, leading to a monotonically decreasing trend of the proposed performance metric (i.e., P) with the increase of PFR, as shown in Figure 5(b). The red line in Figure 5(b) represents the variation of P , which is the average of P_{U-CF} , P_{I-CF} , and P_{H-CF} , with the change of PFR. Based on these experimental results, removing fake reviews can improve the performance of the recommendation algorithms, and this improvement steadily increases with the increasing number of removed fake reviews. Therefore, the proposed metric performance (i.e., P) can be used to evaluate the effectiveness of methods in detecting fake reviews.

Comparison Experiment

As online reviews do not have objective classification information, fake-review detection is an unsupervised problem that can only be addressed by unsupervised learning methods. Since the clustering method is one of the most widely used and classical unsupervised methods [40], this study compares FRD with four unsupervised fake-review detection methods, each based on a different clustering method. These comparative methods include K -means clustering (KM) based detection method [40], density clustering (DC) based detection method [7], hierarchical clustering (HC) based detection method [4], and Gaussian clustering (GC) based detection method [11]. The proposed FRD is compared with these four methods regarding fake-review detection. In FRD, fake reviews can be selected from the perspectives of the reviewer, restaurant, and review, respectively represented by $FRD-Rr$, $FRD-Rt$, and $FRD-R$. The basic $FRD-R$ is first selected to compare with

Table 5. Performance Comparison Between *FRD-R* and Existing Detection Methods Based on *KM*, *DC*, *HC*, and *GC*

Methods	<i>FRD-R</i>	<i>KM</i>	<i>DC</i>	<i>HC</i>	<i>GC</i>
<i>P</i>	18.40%	10.87%	12.35%	9.56%	15.32%

clustering-based fake-review detection methods. Top-*N*% is set to 10.35 percent, the estimated rate of Dianping fake reviews in Supplement 2.1.

As shown in Table 5, *FRD-R* outperforms four comparative clustering-based methods. *P* of *FRD-R* is 18.40 percent, indicating that the recommendation performance improves by 18.40 percent after removing its detected fake reviews from the Dianping dataset. Additionally, experiment results show that the four clustering-based detection methods are all effective in fake-review detection since their *P* values are larger than 0.

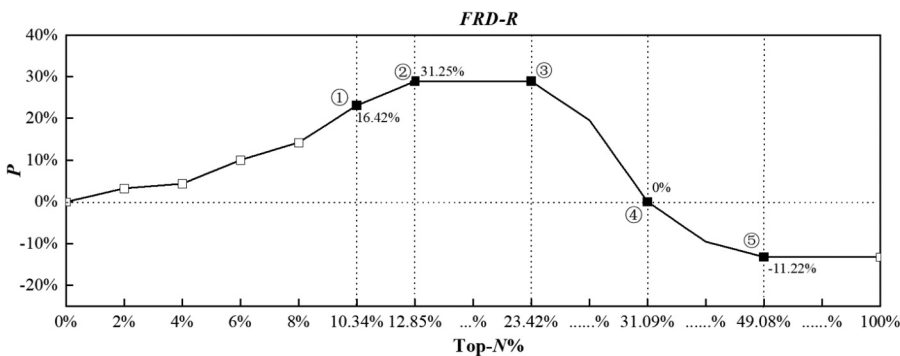
FRD Analysis Experiment

We first conduct experiments to test *FRD-R* performance with different settings of Top-*N*%. Second, we compare *FRD* performance with different strategies for ranking fake reviews. Third, we explore the effects of the fake-review features on the fake index estimation of the proposed detection method.

Analysis of Top-*N*% Setting in *FRD-R*

In this subsection, the performance of *FRD-R*, the basic *FDR* method, is analyzed with different settings of Top-*N*%, a significant parameter determining the number of reviews labeled as fake. In particular, the last step of *FRD-R* involves ranking reviews in descending order based on their fake indexes, and Top-*N*% of reviews are selected as fake reviews. The performance of *FRD-R* is tested with Top-*N*% values ranging from 0 to 100 percent, with increments of 0.01 percent. Due to space limitations, Figure 6 illustrates the key points in the Top-*N*% adjustments.

With an increase of Top-*N*%, *P* reaches its peak value of 31.25 percent when Top-*N*% equals 12.85 percent. The fake review rate estimated based on survey results (i.e., 10.34%) is not the optimal setting for Top-*N*%, as survey research only provides an interval of the fake-review percentage (i.e., [0, 25%] or [25%, 50%]). The average of the interval (i.e., 12.5% and 37.5%) is adopted to estimate the fake review rate.

**Figure 6.** Performance of *FRD-R* Under Different Settings of Top-*N*%

However, with the increase of Top- $N\%$, some authentic reviews are mistakenly labeled as fake. Therefore, when Top- $N\%$ increases from 12.85 percent to 23.42 percent, P remains at 31.25 percent, the highest performance level. Then P begins to drop with the increase of Top- $N\%$. The reason is that the performance of data-driven methods is affected by the volume of data, and better performance is typically achieved with larger amounts of data [28]. P equals 0 percent when the top 31.09 percent of reviews are labeled fake and then reaches its minimum value of -11.22 percent when Top- $N\%$ increases to 49.08 percent. At the point marked as “④” in Figure 6, the positive effect of fake-review detection on recommendations is outweighed by the negative effect of data volume reduction. When the value of Top- $N\%$ exceeds 49.08 percent, MAE_{FD} equals 1 and remains constant, resulting in a constant P value of -11.22 percent.

This experiment confirms the effectiveness of *FRD-R* in detecting fake reviews and the validity of the proposed performance metric P for evaluating detection method performance. The negative effect of data volume reduction would not be an issue when only a small proportion (i.e., less than 20%) of reviews is removed from the dataset [5]. Therefore, P can be used on most online review platforms to evaluate the performance of fake-review detection.

Performance Comparison of Proposed FDR-Based Methods

Compared to *FDR-R*, which serves as the baseline *FDR* method, *FRD-Rr* and *FRD-Rt* respectively label Top- $N\%$ of reviews for each reviewer and each restaurant as fake. Their detection performance is tested with a Top- $N\%$ from 0 to 100 percent with a step of 0.01 percent. The Dianping platform comprises eight reviewer levels, from the lowest, Lv. 1, to the highest, Lv. 8. Reviewers with Lv.5 to Lv.8 are considered high-level reviewers. Considering that the survey results indicate that most fake reviews are sent from high-level reviewers, an extension of *FRD-Rr*, namely, *FRD-Rr (H)*, is developed, which only labels the Top- $N\%$ of reviews from high-level reviewers as fake.

Based on the experimental results, we observed similar inverted-U-shape curves in the performance of *FRD-Rt*, *FRD-Rr*, and *FRD-Rr (H)*, like the curve in Figure 6 for *FRD-R*. These curves have five key points, and the Top- $N\%$ values of these key points are presented in Table 6. While the performance of *FRD-Rt* and *FRD-Rr* is no better than that of the baseline *FRD-R*, *FRD-Rr (H)* outperforms the others significantly. Specifically, the recommendation performance improves by 42.89% when fake reviews detected by *FRD-Rr (H)* are removed from the Dianping dataset. Since the survey results suggest that most fake reviews come from high-level reviewers, authentic reviews from low-level reviewers may be mistakenly labeled as fake by *FRD-Rr* and *FRD-Rt*. In other words, *FRD-Rr (H)* captures more accurate features of review manipulation, which leads to its superior performance in detecting fake reviews in the Dianping dataset compared to the other *FRD* methods.

Table 6. Performance Comparison of *FRD*-Based Methods (P and Top- $N\%$)

Points Method	Label	①	②	③	④	⑤
<i>FRD-R</i> (Baseline)	P	16.42%	31.25%	31.25%	0%	-11.22%
	(Top- $N\%$)	(10.34%)	(12.85%)	(23.42%)	(31.09%)	(49.08%)
<i>FRD-Rt</i>	P	14.28%	26.94%	26.94%	0%	-11.22%
	(Top- $N\%$)	(10.34%)	(14.56%)	(23.76%)	(36.93%)	(50.91%)
<i>FRD-Rr</i>	P	21.32%	28.83%	28.83%	0%	-11.22%
	(Top- $N\%$)	(10.34%)	(13.48%)	(25.68%)	(33.81%)	(49.98%)
<i>FRD-Rr (H)</i>	P	20.91%	42.89%	42.89%	0%	-10.01%
	(Top- $N\%$)	(10.34%)	(15.63%)	(26.67%)	(40.87%)	(100%)

Effect of Fake-Review Features on FRD Performance

Estimating the fake index is a crucial step in the proposed *FRD*. The fake index is composed of various review features, including pictures (i.e., FI_{ij}^{pic}), text (i.e., FI_{ij}^{text}), ratings (i.e., FI_{ij}^{rating}), and reviewers (i.e., L_i), as shown in Equation 7 (i.e., $FI_{ij} = L_i * (1 + FI_{ij}^{pic}) * (1 + FI_{ij}^{text}) * (1 + FI_{ij}^{rating})$). Different combinations are used to estimate the fake index and observe changes in the detection performance of the proposed *FRD*-based methods to investigate the effects of these features on fake-review detection. Top- $N\%$ values are set to the optimal ones obtained in the preceding subsection, and the detection performance evaluated in that section is used as the benchmark.

Table 7 presents the experimental results, demonstrating that the performance relationships of the four *FRD*-based methods remain unchanged regardless of the fake-index estimations, consistent with their relationships in Table 6. Specifically, *FRD-Rr (H)* outperforms the other methods, while *FRD-Rt* and *FRD-Rr* perform worse than *FRD-R*. These results also reinforce the robustness of the earlier results.

The performance rankings of the eight fake-index estimations remain the same across four *FDR* methods, reinforcing the following interpretation of the results. The reviewer level (i.e., L) is the most significant factor in fake index estimation, as the detection performance is greatly affected without it. Including picture features leads to better detection performance than using only text or rating features, and text features are more significant than rating features in the estimation. Since individuals cannot take the same pictures, identical pictures in different reviews may be provided as materials by unscrupulous restaurant owners. Therefore, the appearance of the same pictures in different reviews is a crucial factor in detecting fake reviews. In comparison, text and rating features in fake reviews may also exist in authentic reviews. For example, though most fake reviews tend to have 5-star ratings, a review containing a 5-star rating is not necessarily fraudulent. In fake-review detection, ratings and text play an auxiliary role compared to the reviewer level and pictures. Thus, the significant relationship of review features in detecting fake reviews on Dianping is $L > FI^{image} > FI^{text} > FI^{rating}$.

We conduct weight experiments to further explore the significance of fake-review features in the detection. As FI^{pic} and L_i are positive integers, we first normalize them into the same value range (i.e., $[0, 1]$) as FI_{ij}^{text} and FI_{ij}^{rating} . The fake index can be determined by

Table 7. Effect of Fake-Review Features on *FRD*-Based Methods

Methods					
Fake-index estimation	<i>FRD-R</i>	<i>FRD-Rr</i>	<i>FRD-Rt</i>	<i>FRD-Rr (H)</i>	Rank
$L_i * [(1 + FI_{ij}^{image}) * (1 + FI_{ij}^{text}) * (1 + FI_{ij}^{rating})]$ (Benchmark)	33.12%	26.94%	28.83%	42.89%	1
$[(1 + FI_{ij}^{image}) * (1 + FI_{ij}^{text}) * (1 + FI_{ij}^{rating})]$	5.32%	3.24%	2.85%	9.72%	8
$L_i * [(1 + FI_{ij}^{image}) * (1 + FI_{ij}^{text})]$	32.01%	25.98%	26.82%	39.01%	2
$L_i * [(1 + FI_{ij}^{image}) * (1 + FI_{ij}^{rating})]$	29.84%	23.01%	22.89	36.05	3
$L_i * [(1 + FI_{ij}^{text}) * (1 + FI_{ij}^{rating})]$	13.65%	13.11%	10.87%	14.56%	5
$L_i * (1 + FI_{ij}^{rating})$	8.92%	7.83%	6.01%	11.24%	7
$L_i * (1 + FI_{ij}^{text})$	11.08%	8.55%	9.24%	13.78%	6
$L_i * (1 + FI_{ij}^{image})$	23.84%	20.01%	21.9%	27.26%	4

any combination of FI_{ij}^{pic} , FI_{ij}^{text} , FI_{ij}^{rating} , and L_i . Thus, there are 15 (i.e., $c_4^1 + c_4^2 + c_4^3 + c_4^4$) fake-review index estimations based on the selection of different fake-review features. The weight of the fake-review feature ranges from 0 to 1, and we adjust these with a step of 0.05. Then we evaluate the performance of the *FDR*-based methods using different fake index estimations. Notably, if more than one feature is considered in the fake index estimation, the sum of their weights must equal 1. Due to space limitations, the weight settings for the 15 fake index estimations that yield the best performance for the *FDR*-based methods are presented in Table 8, where the bold numbers in the first column are weight values.

The experimental results first show that all four fake-review features are significant in fake-review detection, and the significance relationship of them is $L > FI^{image} > FI^{text} > FI^{rating}$, which is consistent with the experimental results in Table 7. Notably, $0.45L_i + 0.35FI_{ij}^{image} + 0.15FI_{ij}^{text} + 0.05FI_{ij}^{rating}$ outperforms Equation 7 with equal weights (i.e., $L_i + FI_{ij}^{image} + FI_{ij}^{text} + FI_{ij}^{rating}$) in fake-review detection. Therefore, we recommend that the target platform use this estimation for their fake-review detection.

Further Discussion of Text Similarity Estimation

To justify the use of Equation 5, $Ts_{p,q}^t = Jc_{p,q}^D * (1 + Cs_{p,q}^T)$, in proposed detection methods *FDR-R*, *FRD-Rt*, *FRD-Rr*, and *FRD-Rr (H)*, comparative experiments of these four methods are conducted with four text similarity estimations as shown in the following:

$$Ts_{p,q}^t = \begin{cases} Jc_{p,q}^D & (a) \\ Cs_{p,q}^T & (b) \\ Cs_{p,q}^T * (1 + Jc_{p,q}^{D, Norm}) & (c) \\ \alpha * Jc_{p,q}^{D, Norm} + (1 - \alpha) * Cs_{p,q}^T & (d) \end{cases} \quad (10)$$

In Equations 10(a) and 10(b), we estimate text similarity solely based on Jc and Cs , respectively. Equation 10(c) stands in stark opposition to Equation 5. Within Equation 10 (c), Cs assumes the position of the “main factor,” while Jc operates as the “enhancement factor.” Equation 10(d) presents a linear combination of these two factors. Within Equation 10(d), we introduce the parameter α , falling into the interval of $[0, 1]$, to adjust the relative significance of Jc and Cs in the estimation of text similarity. It is imperative to acknowledge that Jc , being a positive integer greater than 0, cannot be directly applied as the “enhancement factor” in Equation 10(c) or as part of the weighted addition in Equation 10 (d). Given this constraint, a normalization process is taken, which transforms $Jc_{p,q}^D$ into the interval $[0, 1]$. This normalized representation of $Jc_{p,q}^D$ is denoted as $Jc_{p,q}^{D, Norm}$.

This section analyzes text similarity estimations by observing fake-review detection performances of *FDR-R*, *FRD-Rt*, *FRD-Rr*, and *FRD-Rr (H)* on the Dianping dataset. In Table 6, the Top- $N\%$ values of these four methods are set as 12.85 percent, 14.56 percent, 13.48 percent, and 15.63 percent, respectively. Under these settings, they demonstrate the most effective detection performance, accompanied by corresponding P of 31.25 percent, 26.94 percent, 28.83 percent, and 42.89 percent. These four P values are regarded as the baseline in Table 9. Additionally, considering the presence of a weight parameter α in Equation 10(d), α is varied from 0 to 1 with a step of 0.05 to assess the performance of the four methods. The optimal performance is observed when α equals 0.95.

Table 8. Effect of Weight Combinations of Fake-Review Features on the Detection Performance

Methods					
Fake index estimation	<i>FRD-P</i>	<i>FRD-Rr</i>	<i>FRD-Rw</i>	<i>FRD-Rr (H)</i>	<i>Rank</i>
$0.45L_i + 0.35F_{i,j}^{image} + 0.15F_{i,j}^{text} + 0.05F_{i,j}^{rating}$	40.81%	27.40%	28.99%	50.12%	1
$0.50F_{i,j}^{image} + 0.40F_{i,j}^{text} + 0.10F_{i,j}^{rating}$	12.99%	7.09%	10.03%	23.11%	9
$0.60L_i + 0.25F_{i,j}^{image} + 0.15F_{i,j}^{text}$	34.89%	25.93%	24.87%	44.09%	2
$0.55L_i + 0.40F_{i,j}^{image} + 0.05F_{i,j}^{rating}$	31.42%	24.80%	22.33%	39.92%	3
$0.75L_i + 0.20F_{i,j}^{text} + 0.05F_{i,j}^{rating}$	24.14%	19.03%	18.42%	33.99%	5
$0.95L_i + 0.05F_{i,j}^{rating}$	15.50%	10.40%	13.02%	25.88%	7
$0.70L_i + 0.30F_{i,j}^{text}$	19.92%	16.32%	15.55%	29.32%	6
$0.60L_i + 0.40F_{i,j}^{image}$	30.10%	22.78%	19.09%	36.00%	4
$0.75F_{i,j}^{image} + 0.25F_{i,j}^{text}$	11.03%	6.45%	9.88%	20.50%	10
$0.90F_{i,j}^{image} + 0.10F_{i,j}^{rating}$	9.90%	5.66%	8.23%	18.65%	11
$0.70F_{i,j}^{text} + 0.30F_{i,j}^{rating}$	6.01%	3.99%	5.82%	11.20%	13
L_i	13.31%	8.52%	12.99%	23.67%	8
$F_{i,j}^{image}$	8.07%	4.36%	7.25%	14.88%	12
$F_{i,j}^{text}$	4.92%	3.08%	4.12%	7.76%	14
$F_{i,j}^{rating}$	3.93%	2.81%	1.99%	5.90%	15

Table 9. Effect of Text Similarity Estimations on *FRD*-Based Methods

Methods					
Text similarity estimation	<i>FRD-R</i>	<i>FRD-Rr</i>	<i>FRD-Rt</i>	<i>FRD-Rr (H)</i>	<i>Rank</i>
Baseline: $Ts_{p,q}^t = Jc_{p,q}^D * (1 + Cs_{p,q}^T)$	31.25%	26.94%	28.83%	42.89%	1
10(a): $Jc_{p,q}^D$	25.31%	18.55%	21.63%	34.51%	3
10(b): $Cs_{p,q}^T$	9.34%	6.92%	7.76%	11.25%	5
10(c): $Cs_{p,q}^T * (1 + Jc_{p,q}^{D, Norm})$	12.33%	9.82%	10.54%	15.69%	4
10(d): $\alpha * Jc_{p,q}^{D, Norm} + (1 - \alpha) * Cs_{p,q}^T (\alpha = 0.95)$	28.34%	21.09%	24.22%	39.21%	2

As shown in Table 9, in comparison to the other text similarity estimations, both Equations 10(c) and 10(b) display notably inferior performance, with 10(c) showing a slight edge over 10(b). Regarding Equations 10(a) and 10(d), their performance is comparable but falls behind the baseline. Within the scope of these two estimations, 10(d) outperforms 10(a), and when α is set to 0.95 in 10(d), it demonstrates the optimal performance. These results suggest that in text similarity estimation, the evaluated dish similarity (i.e., Jc) should carry more significance than the evaluated content similarity (i.e., Cs). Nevertheless, the significance of Cs should not be disregarded, as demonstrated by the superior performance of α set to 0.95 and the performance of both 10(d) and the baseline over 10(a). Additionally, the “enhanced” manner of combining Jc and Cs within the baseline proves to be the most effective, outperforming their linear combination (i.e., 10(d)).

Conclusion

Detecting fake reviews is crucial in preserving the reliability and credibility of online platforms and ensuring that consumers receive genuine and trustworthy information while making purchasing decisions. The similarities between fake and real reviews today, combined with the unsupervised nature of fake-review detection, present significant challenges for addressing this issue. Therefore, this research proposes an unsupervised method for detecting fake reviews and a performance metric based on recommendations. The experiments assess the feasibility of the proposed

metric using Yelp data and evaluate the performance of the proposed method through a case study on Dianping. According to the experiment results, first, the feasibility of the proposed metric in evaluating fake-review detection performance is verified since the accuracy of recommendations increases monotonically with the removal of fake reviews. Second, the proposed detection method outperforms state-of-the-art unsupervised detection methods. Third, most fake reviewers are high-level reviewers, and the significance of fake-review features is ranked from high to low as reviewer level, image, text, and rating.

However, one of the most significant advantages of our proposed method is that hard-to-find fake-review features can be accurately extracted and analyzed through survey research, which nonetheless presents the main disadvantage of this method. As we combine fake-review features identified in existing studies and those analyzed through our survey results, the specific fake-index estimation cannot be generalized and applied to other platforms, such as Amazon, Tamil, or Yelp. In addition, survey responses cannot always reflect the participants' "true" preferences. Nevertheless, the proposed detection method can be only based on the fake-review features identified in existing studies, and it could have high generality but potentially limited effectiveness.

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Disclosure statement

No potential conflicts of interest are reported by the authors(s).

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References

1. Adomavicius, G.; and Zhang, J. Classification, Ranking, and Top-K stability of recommendation algorithms. *INFORMS Journal on Computing*, 28, 1 (2016), 129–147.
2. Barbado, R.; Araque, O.; and Iglesias, C.A. A framework for fake review detection in online consumer electronics retailers. *Information Processing & Management*, 56, 4 (2019), 1234–1244.
3. Berry, M.W.; Mohamed, A.; and Yap, B.W. *Supervised and Unsupervised Learning for Data Science*. New York: Springer, 2019.
4. Birim, Ş.Ö.; Kazancoglu, I.; Kumar Mangla, S.; Kahraman, A.; Kumar, S.; and Kazancoglu, Y. Detecting fake reviews through topic modelling. *Journal of Business Research*, 149 (2022), 884–900.
5. Cecchini, M.; Aytug, H.; Koehler, G.J.; and Pathak, P. Detecting management fraud in public companies. *Management Science*, 56, 7 (2010), 1146–1160.
6. Dash, A.; Zhang, D.; and Zhou, L. Personalized ranking of online reviews based on consumer preferences in product features. *International Journal of Electronic Commerce*, 25, 1 (2021), 29–50.
7. Duan, L.; Xu, L.; Guo, F.; Lee, J.; and Yan, B. A local-density based spatial clustering algorithm with noise. *Information Systems*, 32, 7 (2007), 978–986.
8. Esposito, M.; Damiano, E.; Minutolo, A.; De Pietro, G.; and Fujita, H. Hybrid query expansion using lexical resources and word embeddings for sentence retrieval in question answering. *Information Sciences*, 514 (2020), 88–105.
9. Fang, L. The effects of online review platforms on restaurant revenue, consumer learning, and welfare. *Management Science*, 68, 11 (2022), 8116–8143.
10. Filieri, R.; Alguezaui, S.; and McLeay, F. Why do travelers trust TripAdvisor? Antecedents of trust towards consumer-generated media and its influence on recommendation adoption and word of mouth. *Tourism Management*, 51 (2015), 174–185.
11. Gebru, I.D.; Alameda-Pineda, X.; Forbes, F.; and Horaud, R. EM algorithms for weighted-data clustering with application to audio-visual scene analysis. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 38, 12 (2016), 2402–2415.
12. He, J.; Fang, X.; Liu, H.; and Li, X. Mobile app recommendation an involvement-enhanced approach. *MIS Quarterly*, 43, 3 (2019).
13. He, S.; Hollenbeck, B.; and Proserpio, D. The market for fake reviews. *Marketing Science*, 41, 5 (2022), 896–921.
14. Hu, N.; Liu, L.; and Sambamurthy, V. Fraud detection in online consumer reviews. *Decision Support Systems*, 50, 3 (2011), 614–626.
15. Huang, G.; Song, S.; Gupta, J.N.D.; and Wu, C. Semi-supervised and unsupervised extreme Learning Machines. *IEEE Transactions on Cybernetics*, 44, 12 (2014), 2405–2417.
16. Iwata, T.; Hirao, T.; and Ueda, N. Topic models for unsupervised cluster matching. *IEEE Transactions on Knowledge and Data Engineering*, 30, 4 (2018), 786–795.
17. Jiang, Y.; Shang, J.; and Liu, Y. Maximizing customer satisfaction through an online recommendation system: A novel associative classification model. *Decision Support Systems*, 48, 3 (2010), 470–479.
18. Jordan, M.I.; and Mitchell, T.M. Machine learning: Trends, perspectives, and prospects. *Science*, 349, 6245 (2015), 255–260.
19. Khansa, L.; Zobel, C.W.; and Goicochea, G. Creating a taxonomy for mobile commerce innovations using social network and cluster analyses. *International Journal of Electronic Commerce*, 16, 4 (2012), 19–52.
20. Knuth, T.; and Ahrholdt, D.C. Consumer fraud in online shopping: detecting risk indicators through data mining. *International Journal of Electronic Commerce*, 26, 3 (2022), 388–411.
21. Kothari, R.; and Jain, V. Learning from labeled and unlabeled data using a minimal number of queries. *IEEE Transactions on Neural Networks*, 14, 6 (2003), 1496–1505.
22. Kumar, A.; Gopal, R.D.; Shankar, R.; and Tan, K.H. Fraudulent review detection model focusing on emotional expressions and explicit aspects: Investigating the potential of feature engineering. *Decision Support Systems*, 155 (2022), 113728.

23. Kumar, N.; Venugopal, D.; Qiu, L.; and Kumar, S. Detecting review manipulation on online platforms with hierarchical supervised learning. *Journal of Management Information Systems*, 35, 1 (2018), 350–380.
24. Kumar, N.; Venugopal, D.; Qiu, L.F.; and Kumar, S. Detecting anomalous online reviewers: An unsupervised approach using mixture models. *Journal of Management Information Systems*, 36, 4 (2019), 1313–1346.
25. Kumar, R.; Mukherjee, S.; and Rana, N.P. Exploring latent characteristics of fake reviews and their intermediary role in persuading buying decisions. *Information Systems Frontiers* (2023).
26. Lappas, T.; Sabnis, G.; and Valkanas, G. The impact of fake reviews on online visibility: A vulnerability assessment of the hotel industry. *Information Systems Research*, 27, 4 (2016), 940–961.
27. Lau, R.Y.K.; Liao, S.Y.; Kwok, R.C.-W.; Xu, K.; Xia, Y.; and Li, Y. Text mining and probabilistic language modeling for online review spam detection. *ACM Transactions on Information Systems*, 2, 4 (2012), 1–30.
28. Li, L.; Chen, J.; and Raghunathan, S. Recommender system rethink: Implications for an electronic marketplace with competing manufacturers. *Information Systems Research*, 29, 4 (2018), 1003–1023.
29. Li, Y.; Zhang, Z.; Pedersen, S.; Liu, X.; and Zhang, Z. The influence of relative popularity on negative fake reviews: A case study on restaurant reviews. *Journal of Business Research*, 162 (2023), 113895.
30. Liang, N.; Biros, D.P.; and Luse, A. An empirical validation of malicious insider characteristics. *Journal of Management Information Systems*, 33, 2 (2016), 361–392.
31. Liu, R.; Liu, R.; Pugliese, A.; and Subrahmanian, V.S. STARS: Defending against sockpuppet-based targeted attacks on reviewing systems. *ACM Transactions on Intelligent Systems and Technology*, 11, 5 (2020), 1–25.
32. Luca, M.; and Zervas, G. Fake it till you make it: Reputation, competition, and Yelp review fraud. *Management Science*, 62, 12 (2016), 3412–3427.
33. Mukherjee, A.; Kumar, A.; Liu, B.; Wang, J.; Hsu, M.; Castellanos, M.; and Ghosh, R. Spotting opinion spammers using behavioral footprints. *Proceedings of the 19th ACM SIGKDD international conference on Knowledge discovery and data mining*, Chicago, IL: Association for Computing Machinery, 2013, pp. 632–640.
34. Munzel, A. Assisting consumers in detecting fake reviews: The role of identity information disclosure and consensus. *Journal of Retailing and Consumer Services*, 32 (2016), 96–108.
35. Nguyen, T.; and Hsu, P.F. More personalized, more useful? Reinvestigating recommendation mechanisms in e-commerce. *International Journal of Electronic Commerce*, 26, 1 (2022), 90–122.
36. Pan, Y.; Chen, S.; Wu, D.; and Dolgui, A. CF-NN: A novel decision support model for borrower identification on the peer-to-peer lending platform. *International Journal of Production Research* (2020), 1–12.
37. Pan, Y.; and Wu, D. A novel recommendation model for online-to-offline service based on the customer network and service location. *Journal of Management Information Systems*, 37, 2 (2020), 563–593.
38. Pan, Y.; Wu, D.; and Olson, D.L. Online to offline (O2O) service recommendation method based on multi-dimensional similarity measurement. *Decision Support Systems*, 103 (2017), 1–8.
39. Raccuglia, P.; Elbert, K.C.; Adler, P.D.F.; Falk, C.; Wenny, M.B.; Mollo, A.; Zeller, M.; Friedler, S.A.; Schrier, J.; and Norquist, A.J. Machine-learning-assisted materials discovery using failed experiments. *Nature*, 533, 7601 (2016), 73–76.
40. Rathore, P.; Soni, J.; Prabakar, N.; Palaniswami, M.; and Santi, P. Identifying groups of fake reviewers using a semisupervised approach. *IEEE Transactions on Computational Social Systems*, 8, 6 (2021), 1369–1378.
41. Rong, X.; Yi, C.; and Tian, Y. Unambiguous scene text segmentation with referring expression comprehension. *IEEE Transactions on Image Processing*, 29 (2020), 591–601.
42. Roy, A.; Qureshi, S.; Pande, K.; Nair, D.; Gairola, K.; Jain, P.; Singh, S.; Sharma, K.; Jagadale, A.; Lin, Y.; Sharma, S.; Gotety, R.; Zhang, Y.; Tang, J.; Mehta, T.; Sindhanuru, H.; Okafor, N.;

- Das, S.; Gopal, C.N.; Rudraraju, S.B.; and Kakarlapudi, A.V. Performance comparison of machine learning platforms. *INFORMS Journal on Computing*, 31, 2 (2019), 207–225.
43. Satia Budhi, G.; Chiong, R.; Wang, Z.; and Dhakal, S. Using a hybrid content-based and behaviour-based featuring approach in a parallel environment to detect fake reviews. *Electronic Commerce Research and Applications*, 47 (2021), 101048.
 44. Schuckert, M.; Liu, X.; and Law, R. Insights into suspicious online ratings: Direct evidence from TripAdvisor. *Asia Pacific Journal of Tourism Research*, 21, 3 (2016), 259–272.
 45. Shan, G.; Zhou, L.; and Zhang, D. From conflicts and confusion to doubts: Examining review inconsistency for fake review detection. *Decision Support Systems*, 144 (2021), 113513.
 46. Song, Y.; Wang, L.; Zhang, Z.; and Hikkerova, L. Do fake reviews promote consumers' purchase intention? *Journal of Business Research*, 164 (2023), 113971.
 47. Wang, G.; Wong, K.W.; and Lu, J. AUC-based extreme learning machines for supervised and semi-supervised imbalanced classification. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 51, 12 (2021), 7919–7930.
 48. Wang, W.; Li, F.; and Ng, M.K. Structural similarity-based nonlocal variational models for image restoration. *IEEE Transactions on Image Processing*, 28, 9 (2019), 4260–4272.
 49. Wu, Y.; Ngai, E.W.T.; Wu, P.; and Wu, C. Fake online reviews: Literature review, synthesis, and directions for future research. *Decision Support Systems*, 132 (2020), 113280.
 50. Xu, Y.; Yang, Y.; Han, J.; Wang, E.; Ming, J.; and Xiong, H. Slandorous user detection with modified recurrent neural networks in recommender system. *Information Sciences*, 505 (2019), 265–281.
 51. Zhang, D.; Zhou, L.; Kehoe, J.L.; and Kilic, I.Y. What online reviewer behaviors really matter? Effects of verbal and nonverbal behaviors on detection of fake online reviews. *Journal of Management Information Systems*, 33, 2 (2016), 456–481.
 52. Zhang, W.; Xie, R.; Wang, Q.; Yang, Y.; and Li, J. A novel approach for fraudulent reviewer detection based on weighted topic modelling and nearest neighbors with asymmetric Kullback–Leibler divergence. *Decision Support Systems*, 157 (2022), 113765.
 53. Zhang, Z.P.; Kudo, Y.; and Murai, T. Neighbor selection for user-based collaborative filtering using covering-based rough sets. *Annals of Operations Research*, 256, 2 (2017), 359–374.